PCMI lecture notes on random matrix theory

Terence Tao · 7 June 2017

https://terrytao.wordpress.com/2017/06/07/pcmi-lecture-notes-on-random-matrix-theory/

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In July I will be spending a week at Park City, being one of the mini-course lecturers in the Graduate Summer School component of the Park City Summer Session on random matrices. I have chosen to give some lectures on least singular values of random matrices, the circular law, and the Lindeberg exchange method in random matrix theory; this is a slightly different set of topics than I had initially advertised (which was instead about the Lindeberg exchange method and the local relaxation flow method), but after consulting with the other mini-course lecturers I felt that this would be a more complementary set of topics. I have uploaded an draft of my lecture notes (some portion of which is derived from my monograph on the subject); as always, comments and corrections are welcome.

[Update, June 23: notes revised and reformatted to PCMI format. -T.]

[Update, Mar 19 2018: further revision. -T.]

Comments (7)

 **Anonymous** · 9 June 2017

Will there be any videos posted online of the lectures?

 **Tom Copeland** · 12 June 2017

No connections to free probability theory?

 **Nick Cook** · 15 June 2017

Connections to free probability will be the focus of the lectures by Shlyakhtenko.

 **Terry Cheng** · 17 June 2017

Will there be any reports online?

 **Danilo** · 21 June 2017

Even though I do not understand anything, I want to watch the lecture online

 **Julian** · 29 June 2017

This is probably a rather weird question, but what is the bibliography style and/or package used in the lecture notes? I find it visually pleasing and very useful with the MR-links and the page back-references
[pcmi.cls, see https://pcmi.ias.edu/author-info -T]

 **Keiji** · 9 July 2017

[correction]
1.1 The epsilon-net argument
– the proof of theorem 1.1.1(Upper bound for operator norm)
P(||Mx|| <= Cn^{1/2} / 2) < P(||Mx|| >= C n^{1/2} / 2) <= exp(-c C^{2} n / 4 + O(n))
Thank you for uploading the lecture note.
[Thanks, this will be corrected in the next revision of the ms. -T.]